

Francisco Miranda Vázquez

| LinkedIn | 55 6663 5106 | francisco.miv4@gmail.com | GitHub |

SUMMARY

Technical developer with a strong foundation in mathematics, statistics, and computational methods - built through a Physics degree at UNAM. Experienced in building data pipelines, web crawlers for technical site analysis, and tracking integrations using Python, SQL, and Google Cloud Platform. Currently working as a Data Analyst at EPA Digital, where I develop crawlers for technical site audits, implement GTM/GA4 tracking solutions, build attribution models in BigQuery, and integrate data across platforms (GA4 - BigQuery - Looker Studio). Deep interest and hands-on experience in AI and machine learning, including neural network development applied to real research problems and AI agent/API integration using the Claude API and Model Context Protocol (MCP). Passionate about automation, system integration, and building practical tools that generate real business impact. Fast learner with proven ability to master new technologies on the job – from zero to productive in tools like GTM, GA4, GCP, and BigQuery within weeks of joining my first professional role.

TECHNICAL SKILLS

- Programming Languages:** Python, R, Julia, JavaScript
- Python libraries:** Pandas, NumPy, SciPy, Matplotlib, Seaborn, TensorFlow, PyTorch, Scikit-Learn, Keras, BeautifulSoup, Requests
- Web & Data Tools:** Google Cloud Platform, Looker Studio, BigQuery, Google Tag Manager, Google Analytics 4, Google Ads, Merchant Center, Firebase
- Tools & Platforms:** Tableau, PowerBI, SQL, Excel, LaTeX, GIT, GitHub
- Languages:** English B2, Italian A2, German A1

PROJECTS

- GTM Audit Analyzer**
Project

June 2026
Python

 - Developed a web crawler to perform full technical audits of websites: GTM/GA4 implementation detection, dataLayer analysis, form mapping, and CTA identification.
 - Generates automated Excel reports with multi-sheet structure summarizing tracking coverage across all crawled pages.
 - Applied in a real marketing analytics context at EPA Digital.
- CNN for Landmark Detection**
Project

December 2024 – Present
Python (transfer learning & fine-tuning)

 - Developed a Convolutional Neural Network for automatic landmark detection and placement on biological structure images.
 - Currently improving the model using transfer learning and fine-tuning with EfficientNet.
 - Presented as an oral talk at the 2025 Latin American Paleontology Congress in Puebla, Mexico. Results to be presented at the 2026 National Paleontology Congress in Baja California Sur, Mexico.
- XML to TPS file converter**
Project

April 2024 – October 2024
Python, R

 - Developed a file format conversion tool for geometric morphometric analysis, transforming XML datasets into TPS format.
 - Presented as an oral talk at the 2024 National Paleontology Congress in Michoacán, Mexico.
- Epifluorescence Microscopy Study – Crystal Defect Characterization**
Instituto de Física UNAM

October 2024
CDMX, Mexico

 - Characterized a crystalline defect in a $RbI : Eu^{2+}$ sample using epifluorescence microscopy, studying the emergence of a grain boundary junction.
 - Presented as an oral talk at the 2024 National Physics Congress in Chihuahua, Mexico.

EXPERIENCE

EPA Digital – Digital Marketing Agency

Data Analyst

January 2026 - Present

CDMX, Mexico

- Developed a web crawler in Python (requests, BeautifulSoup) for automated technical auditing of client websites, detecting GTM/GA4 implementations, dataLayer structure, forms, and CTAs across hundreds of pages.
- Implemented and managed GTM containers, GA4 event tracking, and data layer configurations for multiple clients.
- Built conversion attribution models in BigQuery using SQL, enabling more accurate analysis of marketing performance.
- Designed and maintained data pipelines integrating GA4, Google Ads, and Merchant Center into BigQuery, visualized through Looker Studio dashboards.
- Performed large-scale data cleaning and analysis using Python (Pandas, NumPy) and SQL.
- Worked with Firebase for mobile app tracking and Google Cloud Platform for data infrastructure.

Laboratory Research

Instituto de Física, UNAM

February 2025 - Present

CDMX, Mexico

- Implemented and fine-tuned an EquiformerV2 Transformer-based neural network for interatomic potential generation, working with DFT training data, hyperparameter optimization, and results analysis to enable molecular dynamics simulations of gold chloride, urea, and water clusters.
- Developed the molecular dynamics pipeline using the ASE (Atomic Simulation Environment) library, enabling efficient simulation of gold chloride, urea, and water clusters at significantly reduced computational cost compared to traditional DFT calculations.
- Presented initial results at the 2025 National Physics Congress in Estado de México, Mexico; final results to be presented at the 2026 National Physics Congress in Baja California, Mexico.

EDUCATION

Universidad Nacional Autónoma de México

B.Sc. in Physics (degree completion in progress)

CDMX, Mexico

2020 – 2025

CERTIFICATIONS

Google Skillshop: Google Analytics Certification, Google Ads Measurement Certification, Privacy for Agencies and Partners Certification

Anthropic Academy: Introduction to Agent Skills, Building with the Claude API, Introduction to Model Context Protocol (MCP), Claude Code in Action.

Coursera: Python for Data Science, Data Visualization Fundamentals with Tableau.